

Algebra II Practice: Polynomials, Rational Functions, and Inequalities

Formula and Definition Sheet

Polynomial Functions

A polynomial has the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \quad a_n \neq 0.$$

The **degree** is n , and a_n is the **leading coefficient**. The leading term determines end behavior.

Degree and leading coefficient	As $x \rightarrow -\infty$	As $x \rightarrow \infty$
Even degree, $a_n > 0$	$P(x) \rightarrow \infty$	$P(x) \rightarrow \infty$
Even degree, $a_n < 0$	$P(x) \rightarrow -\infty$	$P(x) \rightarrow -\infty$
Odd degree, $a_n > 0$	$P(x) \rightarrow -\infty$	$P(x) \rightarrow \infty$
Odd degree, $a_n < 0$	$P(x) \rightarrow \infty$	$P(x) \rightarrow -\infty$

If $(x - r)^m$ is a factor, then r is a zero of **multiplicity** m . At a zero of odd multiplicity, the graph crosses the x -axis; at a zero of even multiplicity, it touches and turns. Larger multiplicities flatten the graph near the zero.

Remainder theorem: dividing $P(x)$ by $x - c$ leaves remainder $P(c)$.

Factor theorem: $(x - c)$ is a factor exactly when $P(c) = 0$.

For an integer-coefficient polynomial, possible rational zeros are

$$\pm \frac{\text{factor of constant term}}{\text{factor of leading coefficient}}.$$

Polynomial Inequalities

To solve $P(x) > 0$, $P(x) < 0$, $P(x) \geq 0$, or $P(x) \leq 0$:

1. Move all terms to one side and factor completely.
2. Mark every real zero on a number line.
3. Determine the sign on each interval. The sign changes across odd-multiplicity zeros but not across even-multiplicity zeros.
4. Include zeros for $\leq$ or $\geq$; exclude them for $<$ or $>$.

Rational Functions

A rational function is a quotient

$$R(x) = \frac{P(x)}{Q(x)}, \quad Q(x) \neq 0.$$

Values making the *original* denominator zero are excluded from the domain.

- A canceled denominator factor creates a **hole**. Substitute its x -value into the simplified function to find the hole's y -coordinate.
- An uncanceled denominator zero creates a **vertical asymptote**.
- x -intercepts come from uncanceled numerator zeros. The y -intercept is $R(0)$ when defined.

Horizontal and Oblique Asymptotes

An asymptote describes the graph's long-run behavior; the graph may cross a horizontal or oblique asymptote.

For

$$R(x) = \frac{P(x)}{Q(x)}, \quad n = \deg P, \quad m = \deg Q,$$

compare the degrees after common factors have been canceled:

Degree comparison	End-behavior asymptote
$n < m$	$y = 0$
$n = m$	$y = \frac{\text{leading coefficient of } P}{\text{leading coefficient of } Q}$
$n = m + 1$	oblique (slant) asymptote from polynomial division

A **horizontal asymptote** $y = L$ means

$$R(x) \rightarrow L \quad \text{as } x \rightarrow \infty \quad \text{or as } x \rightarrow -\infty.$$

It describes end behavior, not a boundary that the graph cannot cross.

An **oblique asymptote**, also called a **slant asymptote**, is a line $y = mx + b$ approached by the graph as $x \rightarrow \pm\infty$. Use polynomial division:

$$\frac{P(x)}{Q(x)} = (mx + b) + \frac{\text{remainder}}{Q(x)}.$$

If the remainder term approaches 0, then $y = mx + b$ is the oblique asymptote. For example,

$$\frac{x^2 + 3x + 5}{x - 1} = x + 4 + \frac{9}{x - 1},$$

so the oblique asymptote is $y = x + 4$.

If $n > m + 1$, polynomial division gives a higher-degree polynomial asymptote rather than a horizontal or oblique line.

For rational inequalities, use zeros and vertical asymptotes as critical numbers. Never include a value excluded from the domain.

Instructions

Complete all 20 questions in 60 minutes. For multiple-choice questions, select the best answer. For graphing questions, label intercepts, holes, and asymptotes and show correct end behavior. Give exact values and use interval notation for inequalities.

Problems

1. Which statement describes the end behavior of

$$f(x) = -3x^5 + 2x^3 - 7?$$

- A. Both ends rise.
- B. Both ends fall.
- C. The left end falls and the right end rises.
- D. The left end rises and the right end falls.

2. What is the remainder when $2x^3 - 3x^2 - 5x + 7$ is divided by $x - 2$?

- A. -9
- B. -1
- C. 1
- D. 9

3. Which is the complete factorization of $x^3 - 4x^2 - x + 4$?

- A. $(x - 4)(x - 1)(x + 1)$
- B. $(x + 4)(x - 1)(x + 1)$
- C. $(x - 4)(x^2 + 1)$
- D. $(x - 1)^2(x - 4)$

4. For $P(x) = (x + 2)^2(x - 3)^3$, which statement is true?

- A. The graph crosses at both $x = -2$ and $x = 3$.
- B. The graph touches at both $x = -2$ and $x = 3$.
- C. The graph touches at $x = -2$ and crosses with flattening at $x = 3$.
- D. The graph crosses with flattening at $x = -2$ and touches at $x = 3$.

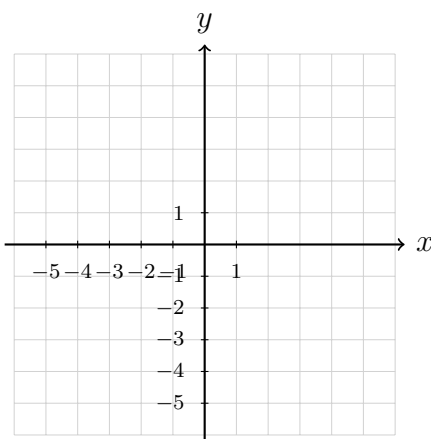
5. Solve $(x - 4)(x + 1)(x - 2) \leq 0$.

- A. $(-\infty, -1] \cup [2, 4]$
- B. $[-1, 2] \cup [4, \infty)$
- C. $(-\infty, -1) \cup (2, 4)$
- D. $[-1, 4]$

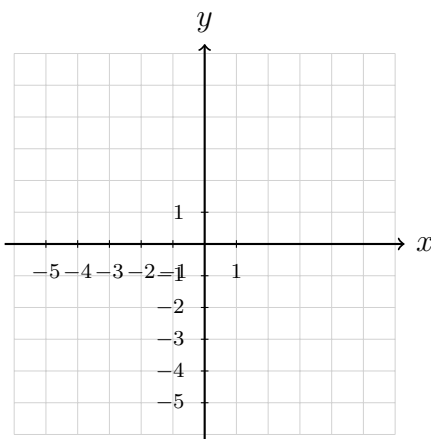
6. Solve $x^4 - 5x^2 + 4 > 0$.

- A. $(-2, -1) \cup (1, 2)$
- B. $(-\infty, -2) \cup (-1, 1) \cup (2, \infty)$
- C. $(-\infty, -1) \cup (1, \infty)$
- D. $(-2, 2)$

7. Graph $y = (x + 3)(x - 1)^2$. Label each intercept and state the multiplicity of each zero.



8. Graph $y = -(x + 2)^2(x - 2)^2$. Label each intercept and show the correct behavior at both zeros.



9. For

$$R(x) = \frac{x^2 - 9}{x^2 - x - 6},$$

which statement is correct?

- A. There is a hole at $(3, \frac{6}{5})$ and a vertical asymptote at $x = -2$.
- B. There is a hole at $(-2, -1)$ and a vertical asymptote at $x = 3$.
- C. There are vertical asymptotes at $x = -2$ and $x = 3$.
- D. There is a hole at $(3, 0)$ and no vertical asymptote.

10. What are the vertical and horizontal asymptotes of

$$R(x) = \frac{2x^2 + x - 3}{x^2 - 4}?$$

- A. $x = \pm 2$ and $y = 0$
- B. $x = \pm 2$ and $y = 2$
- C. $x = 2$ and $y = 2$
- D. $x = -2$ and $y = \frac{1}{2}$

11. What are the intercepts of

$$h(x) = \frac{x - 4}{x^2 - x - 6}?$$

- A. x -intercept $(4, 0)$; y -intercept $(0, \frac{2}{3})$
- B. x -intercept $(-4, 0)$; y -intercept $(0, -\frac{2}{3})$
- C. x -intercepts $(3, 0)$ and $(-2, 0)$; y -intercept $(0, \frac{2}{3})$
- D. No x -intercept; y -intercept $(0, 4)$

12. Solve

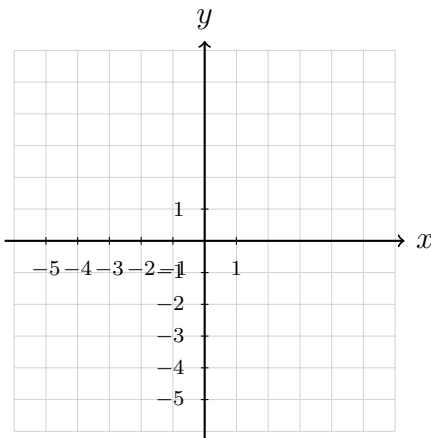
$$\frac{(x - 1)(x + 2)}{x - 3} \geq 0.$$

- A. $(-\infty, -2] \cup [1, 3)$
- B. $[-2, 1] \cup (3, \infty)$
- C. $[-2, 3)$
- D. $(-2, 1) \cup [3, \infty)$

13. Graph

$$y = -\frac{2}{x - 1} + 3.$$

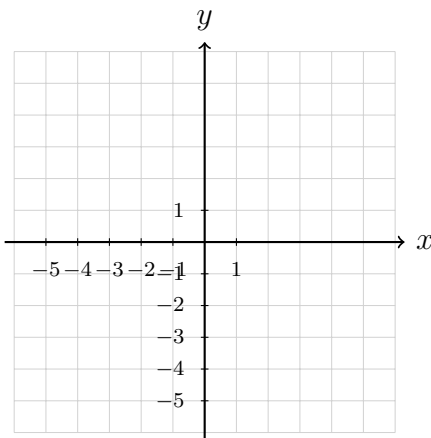
Label both asymptotes and at least two points on the graph.



14. Graph

$$y = \frac{x^2 - 1}{x^2 + x}.$$

Label the hole, vertical asymptote, horizontal asymptote, and intercepts.



15. If $x - 2$ is a factor of $x^3 + kx^2 - 5x + 6$, what is k ?

- A. -4
- B. -1
- C. 1
- D. 4

16. What is the slant asymptote of

$$y = \frac{x^2 + 3x + 5}{x - 1}?$$

- A. $y = x - 4$
- B. $y = x + 1$
- C. $y = x + 4$
- D. $y = 3x + 5$

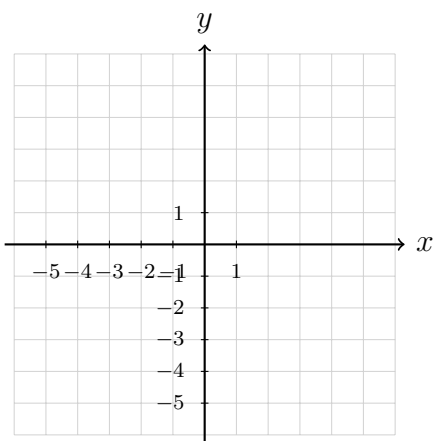
17. Solve $x(x - 2)^2(x + 3) < 0$.

- A. $(-\infty, -3) \cup (0, 2)$
- B. $(-3, 0)$
- C. $(-3, 0) \cup (2, \infty)$
- D. $(-\infty, -3) \cup (0, \infty)$

18. Graph

$$y = -\frac{1}{2}(x + 1)(x - 2)(x - 4).$$

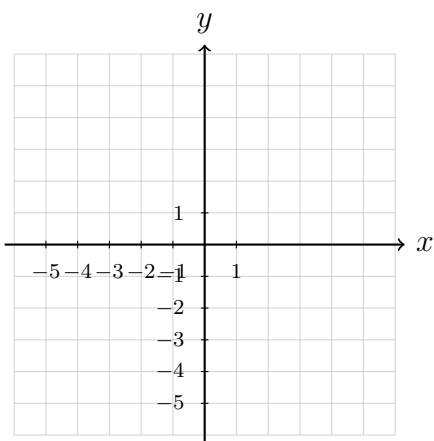
Label all intercepts and show the correct end behavior.



19. Graph

$$y = \frac{x^2 - 4}{x - 1}.$$

Label the vertical and slant asymptotes and all intercepts.



20. A rational function has an x -intercept at $x = 3$, a vertical asymptote at $x = -2$, a horizontal asymptote at $y = 1$, and a hole where $x = 1$. Write one possible function in factored form. Then simplify it, find the hole's y -coordinate, and sketch its graph.

